Mittuniversitetet
MID SWEDEN UNIVERSITY

Försättsblad Prov Original

Kurskod	Provkod	Tentamensdatum
MA080G	T102	2025-06-03
Kursnamn	Matematik GR (B), Kryptografi	
Provnamn	Skriftlig tentamen	
Ort	Sundsvall	
Termin		
Ämne		

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Exam – Cryptography (MA080G)

Mittuniversitetet
Department: IMD
Responsible: Aurélio Menegon

Date: June 3, 2025
Time: 5 hours
No books, notes or symbolic calculators allowed.
Calculator make/model must be stated on the cover page.

Instructions

Answer each subquestion on a separate line. Complete solutions must be provided – reasoning and calculations must be clear and not overly brief. **ALL ANSWERS MUST BE JUSTIFIED.** Grades are based on the achievement of the course objectives.

Grading scale: A (22p), B (18p), C (14p), D (10p), E (9p)

Problems

1. Substitution Ciphers and Vigenère Cipher (4p)

(a) (1p) Decrypt the ciphertext "KHOOR" assuming it was encrypted with a Caesar cipher. Indicate the key used.

(b) (1p) A substitution cipher uses ROT13, where each letter is replaced by the letter 13 positions ahead in the alphabet. Decrypt the message: "GUR DHVPX OEBJA SBK".

(c) (2p) The ciphertext "RIJVS UYVJN" is encrypted with a Vigenère cipher. Suppose the encryption key is "KEY". Determine the corresponding plaintext.

3. Modular Arithmetic, Groups and Euler's Theorem (8p)

(a) (2p) Compute $\phi(90)$ using its prime factorization.

(b) (1p) Is 7 a unit in $\mathbb{Z}/20\mathbb{Z}$? Justify.

(c) (1p) List all generators of $(\mathbb{Z}/9\mathbb{Z}, +)$.

(d) (2p) Compute the order of 2 in $(\mathbb{Z}/9\mathbb{Z})^{\times}$.

(e) (2p) Use Euler's theorem to compute $7^{222} \bmod 40$. Justify all steps.

4. Diffie-Hellman Key Exchange (3p)

Let $p = 29$, $g = 2$, Alice's private key $a = 9$, Bob's private key $b = 14$.

(a) (1p) Compute Bob's public key.

(b) (1p) Compute the shared secret using either party's private key and the other's public key.

(c) (1p) Suppose Eve knows the public parameters $p = 29$, $g = 2$, and the public keys of both Alice and Bob. Would it be possible or easy for Eve to compute the shared secret in this particular situation? Justify your answer based on the principles of the Diffie–Hellman key exchange.

5. ElGamal Encryption (3p)

In the ElGamal encryption scheme, each encryption operation may use a temporary secret number called an *ephemeral key*, denoted by k. This value is chosen randomly for each message encryption and must remain secret. The ephemeral key ensures that encrypting the same message twice produces different ciphertexts, adding semantic security to the system. The ephemeral key is used only during encryption. The recipient does not need to know k to decrypt the message.

Let $p = 113$ and $g = 3$ be public parameters. A user has public key $h = 85$ (computed as $h = g^a \bmod p$ for some private key a), and wishes to send the message $m = 25$. Suppose that the ephemeral key $k = 6$ is randomly selected for this encryption.

(a) (1p) Compute the ciphertext component $c_1 = g^k \bmod p$.

(b) (1p) Compute the ciphertext component $c_2 = m \cdot h^k \bmod p$.

(c) (1p) If the recipient's private key is $a = 21$, decrypt the ciphertext (c_1, c_2) and recover the original message m.

6. RSA Encryption and Decryption (4p)

Let $n = 299$, $e = 5$.

(a) (1p) Factor n and compute $\phi(n)$.

(b) (1p) Compute the private exponent d such that $ed \equiv 1 \mod \phi(n)$.

(c) (1p) Encrypt the message $m = 12$.

(d) (1p) **True or False:** Solving the Discrete Logarithm Problem (DLP) would render the RSA cryptosystem insecure and unusable? Justify your answer briefly.

7. Pollard's ρ Algorithm (2p)

Use Pollard's ρ algorithm to find a nontrivial factor of $n = 221$, using the iteration function $f(x) = x^2 + 1 \bmod 221$ and starting value $x_0 = 2$. Perform the necessary steps of the algorithm, showing the values of x_i, $y_i = f(f(x_{i-1}))$, and $\gcd(|x_i - y_i|, 221)$ at each step.